

SANJAY NJARAKKATTIL

+91-6235221806 ◊ Kadayiruppu, Kerala, India

sanjaynjarakkatil@gmail.com ◊ [linkedin.com/in/sanjay-njarakkatil](https://www.linkedin.com/in/sanjay-njarakkatil) ◊ github.com/san7216

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate (2026, KTU) with 8.35 CGPA, skilled in Python, Java, SQL, and DSA. Hands-on experience in Machine Learning (YOLOv5, CNN), backend development (FastAPI, REST APIs), and web development (React, JavaScript). Seeking entry-level roles in IT Support, QA/Testing, Data Analysis, or Cloud to apply practical project and internship experience in a results-driven team.

EDUCATION

Bachelor of Technology in Computer Science and Engineering	2022 – 2026
Federal Institute of Science and Technology, Kerala Technical University	CGPA: 8.35

Higher Secondary Education	2022
St. Peter's HSS, Kolenchery	98%

Secondary Education (SSLC)	2020
GHSS Kadayiruppu	95%

SKILLS

Languages:	C, Python, Java, SQL
Tools:	Git, GitHub, MySQL, VSCode, Google Colab
Databases:	SQL, PL/SQL
Web:	HTML, CSS, JavaScript, React
Operating System:	Linux Commands, Bash Scripting
Cloud:	AWS(Cloud Foundations), Basic Networking, Basic Linux
Core Concepts:	Data Structures and Algorithms, Object Oriented Programming, DBMS
Behavioral Skills:	Team Collaboration, Problem Solving, Communication, Adaptability

INTERNSHIP

Luminar Technolab – Machine Learning Intern	2025
Implemented multiclass image classification on the Fashion MNIST dataset (60,000 samples, 10 classes) using a Flatten Dense Sequential model with Softmax output, and improved accuracy significantly through data normalization	

CCRC, FISAT – Web Development Intern	2025
Developed a Weather Forecast application that fetches and displays real-time weather data using a public API.	

Qnays LLP – Web Development Intern	2024
Worked on basic web interfaces using HTML, CSS, and JavaScript. Assisted in team projects for web applications and collaborated with developers to implement features.	

STEM Robotics – Python Intern	2024
Completed a 5-day Python internship demonstrating proficiency in Python programming and teamwork. Contributed to projects and initiatives showcasing adaptability and professionalism.	

PROJECTS

Aidify – AI-Powered Accessibility Assistant for Differently-Abled Users

Developed a mobile AI app that detects real-world objects in real time and gives voice guidance to help visually impaired users. Built with a Flutter frontend and Python FastAPI backend using YOLOv5 for object detection, with Supabase for authentication and user data.

Tech Stack: Flutter, Dart, FastAPI, Python, YOLOv5, Supabase, REST APIs

- Implemented a camera-to-API flow where Flutter sends frames to FastAPI, YOLOv5 detects objects, and results are shown with voice feedback.
- Added accessibility features like text-to-speech, multi-language support, and a simple UI, with testing for low-light and real-world detection cases.

DefectPrediction – AI-Powered Vulnerability Detection for C/C++ Code

Developed a full-stack machine learning web application to detect security vulnerabilities in C/C++ source code using hybrid rule-based analysis and deep learning inference.

Integrated a fine-tuned GraphCodeBERT model with manual safety rules and static heuristics to reduce false positives and improve prediction reliability.

Tech Stack: Flask, PyTorch, HuggingFace Transformers, GraphCodeBERT, SHAP, HTML/CSS/JavaScript

- Engineered 5-layer cascade pipeline combining regex-based API checks, memory safety analysis, and GraphCodeBERT ML inference for comprehensive vulnerability detection
- Built Flask REST API with SHAP token attribution and semantic clue extraction for explainable predictions on C/C++ code inputs

HACKATHONS

CodEdge – EDGE+ 4.0, FISAT 2025
Finalist; developed an app for differently-abled individuals.

ICEFOSS24 – National Level 36 Hour Hackathon, FISAT 2024
Participated in National Level 36 Hour Hackathon conducted as part of ICEFOSS’24 organized by FISAT Free Software Cell and ACM Student Chapter FISAT.

CERTIFICATIONS

AWS Cloud Foundations - AWS Academy 2025
Cloud Computing Concepts Using AWS - FISAT 2025
IBM AI Fundamentals 2025
DevOps Fundamentals - IBM Developer Skills Network 2026
Rust Programming - ICEFOSS FISAT 2023
Linux Unhatched(Cisco Networking Academy) 2026
Winter Certification Program (Prompt Engineering)– IBM Skills Build 2025

HOBBIES AND INTERESTS

Coding
Web Development
Cricket
Travelling
Movies

CO-CURRICULAR ACTIVITIES

Volunteered in **MANTRA** (Annual CSE Day), FISAT
Finalist – **CodEdge Hackathon, EDGE+ 4.0, FISAT** (2025)
Participant – **ICEFOSS’24** National Level 36-Hour Hackathon, FISAT (2024)
Active Member of **Student Police Cadet** in School
School **Cricket Team Member**